

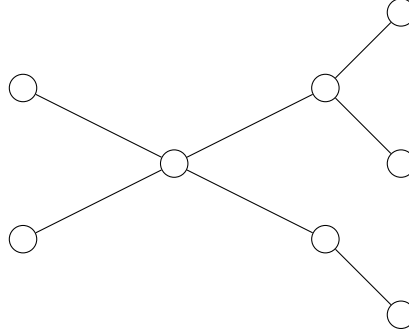
Percolation on the Bethe Lattice

1 Percolation on the Bethe Lattice

The Bethe lattice is an infinite tree-like structure in which each site has exactly z nearest neighbours (coordination number z). Unlike regular lattices, the Bethe lattice contains no loops, which makes different branches statistically independent.

Because of this tree structure, percolation on a Bethe lattice can be understood as a **branching process**. Starting from a site, each occupied site generates new occupied neighbours along the branches of the tree.

1.1 Schematic Structure



Each site connects to z neighbours and the lattice branches outward like a tree. The absence of loops ensures that different branches are independent.

1.2 Number of Sites in a Bethe Lattice

Consider a Bethe lattice with coordination number z . Starting from a central site (generation $g = 0$), the number of sites in successive generations grows exponentially.

Generation structure

$$\begin{aligned} g = 0 & : 1 \\ g = 1 & : z \\ g = 2 & : z(z-1) \\ g = 3 & : z(z-1)^2 \\ & \vdots \\ g & : z(z-1)^{g-1} \end{aligned}$$

Thus the number of sites in generation k is

$$N_k = \begin{cases} 1 & k = 0 \\ z(z-1)^{k-1} & k \geq 1 \end{cases}$$

1.3 Total Number of Sites

The total number of sites up to generation g is

$$N(g) = 1 + \sum_{k=1}^g z(z-1)^{k-1}.$$

Factor z ,

$$N(g) = 1 + z \sum_{k=0}^{g-1} (z-1)^k.$$

Using the geometric series

$$\sum_{k=0}^{g-1} (z-1)^k = \frac{(z-1)^g - 1}{(z-1) - 1} = \frac{(z-1)^g - 1}{z-2},$$

we obtain

$$N(g) = 1 + z \frac{(z-1)^g - 1}{z-2}.$$

Thus

$$\boxed{N(g) = \frac{z(z-1)^g - 2}{z-2}}$$

for $z > 2$.

1.4 Number of Surface Sites

The sites in the outermost generation g form the boundary of the lattice. Their number is

$$N_{\text{surf}} = z(z-1)^{g-1}.$$

1.5 Ratio of Surface Sites to Total Sites

The fraction of sites belonging to the boundary is

$$R(g) = \frac{N_{\text{surf}}}{N(g)} = \frac{z(z-1)^{g-1}}{\frac{z(z-1)^g - 2}{z-2}}.$$

For large g ,

$$N(g) \approx \frac{z}{z-2} (z-1)^g.$$

Therefore

$$R(g) \approx \frac{z(z-1)^{g-1}}{\frac{z}{z-2}(z-1)^g}.$$

Simplifying,

$$R(g) \approx \frac{z-2}{z-1}.$$

Thus in the limit

$$g \rightarrow \infty$$

$$\boxed{R = \frac{z-2}{z-1}}.$$

1.6 Important Property

Unlike ordinary lattices, the fraction of boundary sites in a Bethe lattice remains **finite** even as the system size becomes infinite.

This property is a direct consequence of the exponential growth of the tree structure and plays an important role in percolation on Bethe lattices.

1.7 Strength of the Infinite Cluster

Let

$$P(p)$$

denote the probability that a randomly chosen site belongs to the **infinite cluster**.

To compute this quantity we define an auxiliary probability.

Let

$$Q$$

be the probability that a site is **not connected to infinity** through a particular branch emerging from that site.

1.8 Expression for $P(p)$

A site belongs to the infinite cluster if

- the site itself is occupied
- at least one of its branches connects to infinity

If the probability that a branch fails to reach infinity is Q , then the probability that all z branches fail is

$$Q^z$$

Thus the probability that at least one branch connects to infinity is

$$1 - Q^z$$

Multiplying by the probability that the site is occupied gives

$$P(p) = p(1 - Q^z)$$

However, when following a particular branch we exclude the branch we arrived from. Thus each site effectively has $z - 1$ forward branches.

Therefore

$$P(p) = p(1 - Q^{z-1})$$

For example, for $z = 4$

$$P(p) = p(1 - Q^3).$$

1.9 Recursive Equation for Q

The quantity Q can be determined self-consistently.

A branch does not connect to infinity if

- the first site is empty
- or the site is occupied but none of its $z - 1$ outgoing branches lead to infinity

Thus

$$Q = (1 - p) + pQ^{z-1}.$$

For the case $z = 4$ this becomes

$$Q = (1 - p) + pQ^2.$$

Rearranging,

$$pQ^2 - Q + (1 - p) = 0.$$

1.10 Solution for Q

Solving the quadratic equation gives

$$Q = \frac{1 \pm \sqrt{1 - 4p(1 - p)}}{2p}.$$

Since

$$1 - 4p(1 - p) = (2p - 1)^2$$

we obtain

$$Q = \frac{1 \pm |2p - 1|}{2p}.$$

Thus the two solutions are

$$Q = 1$$

and

$$Q = \frac{1-p}{p}.$$

The physical solution is

$$Q = \begin{cases} 1, & p \leq p_c \\ \frac{1-p}{p}, & p > p_c \end{cases}$$

1.11 Critical Probability

The critical probability occurs when the two solutions coincide. From the general equation

$$Q = (1-p) + pQ^{z-1}$$

the transition occurs when

$$p(z-1) = 1.$$

Thus

$$p_c = \frac{1}{z-1}.$$

For coordination number $z = 4$

$$p_c = \frac{1}{2}.$$

1.12 Strength of the Infinite Cluster

Substituting

$$Q = \frac{1-p}{p}$$

into

$$P(p) = p(1 - Q^3)$$

gives

$$P(p) = p \left(1 - \left(\frac{1-p}{p} \right)^3 \right).$$

Simplifying,

$$P(p) = \frac{p^3 - (1-p)^3}{p^2}.$$

Expanding the numerator,

$$p^3 - (1 - p)^3 = (2p - 1)(p^2 + p(1 - p) + (1 - p)^2).$$

Thus

$$P(p) = \frac{(2p - 1)(p^2 + p(1 - p) + (1 - p)^2)}{p^2}.$$

1.13 Behaviour Near the Critical Point

Near

$$p_c = \frac{1}{2}$$

we expand

$$P(p) \sim (p - p_c).$$

Thus the strength of the infinite cluster grows linearly above the critical probability.

1.14 Mean Finite Cluster Size

The mean size of finite clusters can be obtained using the branching process picture.

Starting from a site,

- the first generation contains z neighbours
- each subsequent generation produces $z - 1$ new neighbours

Each neighbour is occupied with probability p .

Thus the average number of occupied sites in generation k is

$$z[p(z - 1)]^{k-1}.$$

The mean cluster size is therefore

$$S = 1 + zp + zp^2(z - 1) + zp^3(z - 1)^2 + \dots$$

Factoring,

$$S = 1 + zp \sum_{k=0}^{\infty} [p(z - 1)]^k.$$

This is a geometric series,

$$S = 1 + \frac{zp}{1 - p(z - 1)}.$$

Thus the mean cluster size diverges when

$$p(z - 1) = 1.$$

Therefore

$$p_c = \frac{1}{z-1}.$$

This divergence signals the appearance of an infinite cluster.

Algorithm 1 Computation of $Q(p)$ (Branch Failure Probability)

1: **Input:** coordination number z , depth D , occupation probability p , realizations R

2: Initialize fail_count $\leftarrow 0$

3: **for** $r = 1$ to R **do**

4: Simulate one branch:

survives $\leftarrow$ **GrowBranch**(2)

5: **if** survives = 0 **then**

6: fail_count $\leftarrow$ fail_count + 1

7: **end if**

8: **end for**

9: Compute:

$$Q_{\text{num}} = \frac{\text{fail_count}}{R}$$

Algorithm 2 Computation of $P(p)$ (Infinite Cluster Strength)

```
1: Input: coordination number  $z$ , depth  $D$ , occupation probability  $p$ , realizations  $R$ 
2: Initialize success_count  $\leftarrow 0$ 
3: for  $r = 1$  to  $R$  do
4:   Check root occupation
5:   if root is unoccupied (probability  $1 - p$ ) then
6:     continue
7:   end if
8:   branch_success  $\leftarrow 0$ 
9:   for  $b = 1$  to  $z$  do
10:    survives  $\leftarrow$  GrowBranch(2)
11:    if survives = 1 then
12:      branch_success  $\leftarrow 1$ 
13:      break
14:    end if
15:  end for
16:  if branch_success = 1 then
17:    success_count  $\leftarrow$  success_count + 1
18:  end if
19: end for
20: Compute:
```

$$P_{\text{num}} = \frac{\text{success_count}}{R}$$

Algorithm 3 GrowBranch(level)

```
1: if current node is unoccupied (probability  $1 - p$ ) then
2:   return 0 ▷ branch dies
3: end if
4: if level =  $D$  then
5:   return 1 ▷ reaches boundary
6: end if
7: for  $b = 1$  to  $z - 1$  do
8:   child  $\leftarrow$  GrowBranch(level + 1)
9:   if child = 1 then
10:    return 1 ▷ one surviving path is enough
11:   end if
12: end for
13: return 0
```

Algorithm 4 Theoretical Computation of Q and $P(p)$

1: Initialize $Q \leftarrow 1$

2: **repeat**

$$Q_{\text{new}} = 1 - p + pQ^{z-1}$$

3: $Q \leftarrow Q_{\text{new}}$

4: **until** convergence

5: Compute:

$$P(p) = p(1 - Q^z)$$
